Additional file 1

Supplementary methods, tables, and figures

Supplement to:

A Leakage-Aware and Auditable Framework Prioritizes Class I HDAC Inhibitors for Pan-Cancer Transcriptomic Reversal

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Evidence policy. This supplement distinguishes predicted reversal, predicted–measured LINCS concordance, biological context, and structural sensitivity. The dual-stream model has no measurable advantage over the strong conventional fingerprint MLP comparator. Measured reversal is not a viability assay; reversal-associated genes are not asserted direct drug targets; DepMap knockout effects are not compound responses; and docking is not binding validation. The 24 tables below are compact human-readable views. Full, filterable rows and source fields are supplied in Additional file 2.

Evidence freeze: 19 July 2026

Contents

1 Scope, corrections, and candidate policy

The revision uses three completed computational evidence packages: (E1) official-condition measured LINCS comparison and prediction–measurement concordance; (E2) biological reanalysis after candidate roles were frozen, including networks, pathways, and DepMap Public 26Q1; and (E3) controlled zinc-aware docking with crystallographic redocking, a designed zinc-chelation decoy, and receptor sensitivity. A direct external drug-sensitivity package was conditional on candidate-matched public data and was not performed because such data were unavailable for TC-H-106. No new wet-lab experiment is claimed.

The candidate hierarchy was frozen before network, pathway, dependency, and docking interpretation. Mocetinostat is the core computational candidate; NCH-51 is secondary; TC-H-106 is exploratory/original-method-sensitive; RG2833 is prediction-only because no measured LINCS signature was found; and Tianeptinaline/BG-1010 is excluded from primary inference because official identifier resolution exposed a name–structure conflict. Belinostat, PCI-24781, and Panobinostat provide class support; Entinostat and Vorinostat are reference controls.

The revised package removes simulated TCGA survival and stage analyses, unsupported tumor-mutation-burden conclusions, proxy DeepCE/ChemCPA claims, unsupported legacy ablation values, GDSC surrogate "validation," direct-target language for reversed genes, and MMFF94 energy claims as thermodynamic evidence. A pre-correction HDAC holdout is retained only as provenance; all model inference uses the corrected annotation-defined 1,856-profile, 30-structure holdout.

2 Dataset, model scope, and leakage-aware evaluation

Table S1 separates signatures, perturbagen tokens, nominal conditions, canonical structures, and cell lines. This distinction prevents the 55,695 modeling rows from being described as unique compounds or biological replicates.

Table S1: Composition of the quality-controlled LINCS modeling dataset and frozen screening library. Level 5 signatures are not unique compounds or raw biological replicates.

Item	Count	Interpretation
Level 5 signatures	55,695	Modeling rows after quality control
Signature-ID perturbagen tokens	13,314	Includes instance suffixes
Base perturbagen identifiers	11,527	Normalized BRD identifiers
Unique canonical structures	11,445	Valid RDKit structures
Cell lines	65	Across the modeling dataset
Perturbagen-token–cell combinations	39,266	Descriptive combinations
Perturbagen-token–cell–time–dose conditions	52,834	Nominal conditions
Additional signatures beyond nominal conditions	2,861	Plate/batch-level Level 5 entries
Expression genes	12,328	Shared disease/perturbation feature space
Screened canonical compounds	28,477	Frozen screening library; distinct from the modeling structures

Both neural predictors are chemical-structure-only models. They do not receive cell-line identity, dose, or time. Their output is therefore a structure-conditioned central tendency across training conditions, not a condition-specific response. Leave-cell-line evaluation measures agreement with profiles from held-out cellular backgrounds under this formulation; it does not show that the models learned cell-specific response rules.

Table S2: Frozen model, optimization, and interpretation settings.

Item	Frozen specification
Prediction input	Canonical chemical structure only; no cell-line, dose, or time covariates
Target	12,328-gene LINCS Level 5 perturbation vector
Atom stream	43-dimensional element identity; 3 Transformer layers; 512 pooled dimensions; no bond adjacency
Fingerprint stream	1,024-bit radius-2 Morgan fingerprint; 512-dimensional projection
Dual-stream parameters	38,682,152
Fingerprint MLP parameters	28,415,016
Optimization	MSE; AdamW; learning rate 2e-4; weight decay 1e-4; batch 128
Stopping	Maximum 30 epochs; patience 5; checkpoint by validation MSE
Benchmark seeds	Pair split 1–5; other splits 1–3; grouping seed 42
Primary screening metrics	signed wTRS and negative Spearman disease–perturbation correlation
Stability configurations	48 primary; 72 only when legacy wTRS is added as sensitivity

The split audit first checks signature identifiers and then distinguishes exact-structure and scaffold exposure. Zero scaffold overlap is required only for the scaffold split. The corrected annotation-defined HDAC holdout requires zero exact HDAC structure overlap; 11 scaffolds remain shared and are reported rather than mislabeled as leakage.

Table S3: Leakage and structural-overlap audit. Exact-structure overlap is zero for leave-drug, scaffold, and corrected annotation-defined HDAC evaluations; the scaffold column is the directly recorded Murcko overlap.

Split	Train n	Test n	Train struct.	Test struct.	Sig. ovlp.	Scaffold ovlp.
Drug–cell pair	47,376	8,319	10,582	3,827	0	2,489
Leave drug out	47,728	7,967	9,728	1,717	0	686
Leave cell line out	50,520	5,175	11,170	2,826	0	1,914
Scaffold split	47,851	7,844	9,831	1,614	0	0
Corrected annotation-defined HDAC	53,839	1,856	11,415	30	0	11

Table S4: Generalization summary as mean (seed SD). Corrected 1,856-profile HDAC runs replace stale pre-correction summaries.

Split	Model	Seeds	Test	MSE	MAE	R^2	Pearson	Spearman
Corrected annotation-defined HDAC holdout	Dual stream	3	1856	1.326 (0.026)	0.839 (0.005)	0.131 (0.017)	0.378 (0.009)	0.345 (0.010)
Corrected annotation-defined HDAC holdout	Fingerprint MLP	3	1856	1.333 (0.015)	0.841 (0.004)	0.127 (0.010)	0.379 (0.007)	0.344 (0.006)
Leave cell line out	Dual stream	3	5175	0.942 (0.004)	0.714 (0.000)	0.121 (0.003)	0.266 (0.010)	0.243 (0.010)
Leave cell line out	Fingerprint MLP	3	5175	0.941 (0.007)	0.713 (0.002)	0.122 (0.006)	0.272 (0.010)	0.249 (0.010)
Leave drug out	Dual stream	3	7967	0.919 (0.003)	0.717 (0.001)	0.052 (0.003)	0.188 (0.000)	0.166 (0.001)
Leave drug out	Fingerprint MLP	3	7967	0.919 (0.001)	0.716 (0.001)	0.052 (0.001)	0.191 (0.003)	0.170 (0.003)
Drug–cell pair	Dual stream	5	8319	0.882 (0.004)	0.704 (0.003)	0.108 (0.004)	0.259 (0.009)	0.237 (0.009)
Drug–cell pair	Fingerprint MLP	5	8319	0.881 (0.001)	0.703 (0.001)	0.110 (0.002)	0.263 (0.005)	0.241 (0.005)
Scaffold split	Dual stream	3	7844	0.973 (0.000)	0.738 (0.002)	0.062 (0.000)	0.185 (0.006)	0.163 (0.007)
Scaffold split	Fingerprint MLP	3	7844	0.971 (0.002)	0.736 (0.001)	0.064 (0.002)	0.186 (0.001)	0.165 (0.002)

Table S5: All 34 frozen model runs. Runtime includes training and evaluation on the recorded RTX 4090 environment.

Split	Model	Seed	Test	Epoch	Runtime s	MSE	MAE	R^2	Pearson	Spearman
Corrected HDAC	Dual	1	1,856	4	260.5	1.320	0.837	0.135	0.381	0.348
Corrected HDAC	Dual	2	1,856	5	282.7	1.355	0.845	0.112	0.385	0.353
Corrected HDAC	Dual	3	1,856	8	365.0	1.305	0.834	0.145	0.368	0.334
Corrected HDAC	Fingerprint	1	1,856	2	54.1	1.347	0.844	0.118	0.387	0.351
Corrected HDAC	Fingerprint	2	1,856	5	72.5	1.336	0.842	0.125	0.373	0.338
Corrected HDAC	Fingerprint	3	1,856	4	70.3	1.316	0.837	0.138	0.378	0.344
Leave cell	Dual	1	5,175	6	209.5	0.938	0.713	0.125	0.259	0.236
Leave cell	Dual	2	5,175	7	212.4	0.945	0.713	0.118	0.261	0.238
Leave cell	Dual	3	5,175	15	394.8	0.942	0.714	0.121	0.277	0.254
Leave cell	Fingerprint	1	5,175	13	78.9	0.937	0.712	0.125	0.277	0.254
Leave cell	Fingerprint	2	5,175	7	58.4	0.949	0.715	0.115	0.261	0.238
Leave cell	Fingerprint	3	5,175	15	91.5	0.937	0.711	0.126	0.279	0.256
Leave drug	Dual	1	7,967	2	132.4	0.916	0.716	0.055	0.187	0.165
Leave drug	Dual	2	7,967	4	163.5	0.921	0.718	0.049	0.188	0.166
Leave drug	Dual	3	7,967	5	182.0	0.921	0.717	0.050	0.188	0.167
Leave drug	Fingerprint	1	7,967	5	48.5	0.918	0.716	0.053	0.195	0.173
Leave drug	Fingerprint	2	7,967	2	36.7	0.919	0.715	0.052	0.188	0.167
Leave drug	Fingerprint	3	7,967	5	51.7	0.920	0.716	0.051	0.191	0.169
Pair	Dual	1	8,319	13	327.2	0.880	0.702	0.111	0.268	0.246
Pair	Dual	2	8,319	5	191.6	0.888	0.707	0.102	0.248	0.225
Pair	Dual	3	8,319	14	336.4	0.877	0.701	0.113	0.268	0.245
Pair	Dual	4	8,319	8	237.8	0.881	0.704	0.109	0.256	0.233
Pair	Dual	5	8,319	10	278.0	0.885	0.705	0.106	0.258	0.236
Pair	Fingerprint	1	8,319	14	86.7	0.881	0.703	0.109	0.268	0.246
Pair	Fingerprint	2	8,319	11	71.9	0.879	0.702	0.112	0.263	0.241
Pair	Fingerprint	3	8,319	6	55.0	0.883	0.704	0.108	0.254	0.232
Pair	Fingerprint	4	8,319	14	87.5	0.881	0.703	0.110	0.266	0.244
Pair	Fingerprint	5	8,319	10	72.7	0.880	0.702	0.111	0.263	0.240
Scaffold	Dual	1	7,844	2	135.6	0.973	0.735	0.062	0.179	0.156
Scaffold	Dual	2	7,844	4	167.6	0.974	0.739	0.061	0.191	0.170
Scaffold	Dual	3	7,844	3	152.6	0.974	0.739	0.061	0.186	0.163
Scaffold	Fingerprint	1	7,844	6	56.7	0.974	0.736	0.062	0.187	0.165
Scaffold	Fingerprint	2	7,844	3	45.2	0.969	0.735	0.066	0.185	0.163
Scaffold	Fingerprint	3	7,844	3	44.7	0.971	0.736	0.064	0.186	0.166

Table S6: Paired model differences. Positive values favor the dual stream; all intervals include zero except the candidate-specific PCI-24781 comparison reported separately.

Split	Metric	Seeds	Dual advantage	SD	95% low	95% high
Pair	MSE	5	-0.001	0.006	-0.009	0.006
Pair	MAE	5	-0.001	0.003	-0.005	0.003
Pair	R2	5	-0.001	0.006	-0.009	0.006
Pair	Pearson	5	-0.003	0.011	-0.017	0.010
Pair	Spearman	5	-0.003	0.011	-0.018	0.011
Leave drug	MSE	3	-0.001	0.002	-0.006	0.005
Leave drug	MAE	3	-0.002	0.001	-0.005	0.002
Leave drug	R2	3	-0.001	0.002	-0.006	0.005
Leave drug	Pearson	3	-0.003	0.004	-0.013	0.006
Leave drug	Spearman	3	-0.004	0.004	-0.013	0.005
Leave cell	MSE	3	-0.001	0.004	-0.012	0.010
Leave cell	MAE	3	-0.001	0.002	-0.006	0.005
Leave cell	R2	3	-0.001	0.004	-0.011	0.009
Leave cell	Pearson	3	-0.007	0.010	-0.031	0.018
Leave cell	Spearman	3	-0.006	0.010	-0.031	0.018
Scaffold	MSE	3	-0.002	0.003	-0.009	0.004
Scaffold	MAE	3	-0.002	0.003	-0.009	0.004
Scaffold	R2	3	-0.002	0.003	-0.009	0.004
Scaffold	Pearson	3	-0.001	0.007	-0.018	0.016
Scaffold	Spearman	3	-0.002	0.008	-0.022	0.018
Corrected HDAC	MSE	3	0.006	0.023	-0.051	0.064
Corrected HDAC	MAE	3	0.002	0.005	-0.010	0.014
Corrected HDAC	R2	3	0.004	0.015	-0.034	0.042
Corrected HDAC	Pearson	3	-0.001	0.011	-0.030	0.027
Corrected HDAC	Spearman	3	0.001	0.013	-0.030	0.032

Across generalization settings, the strong conventional fingerprint MLP comparator was slightly better on average in the pair, leave-drug, leave-cell-line, and scaffold evaluations. The corrected HDAC holdout was mixed by metric and every paired interval included zero. These tables replace all proxy state-of-the-art comparisons and show no measurable gain from the dual-stream architecture.

3 Candidate identity, condition coverage, and corrected holdout

Table S7: Candidate identity, BRD identifier, and frozen evidence role. Tianeptinaline/BG-1010 is retained only to document the identity conflict.

Candidate	BRD ID	Frozen role	HDAC annotated	Name conflict	Interpretation
Mocetinostat	BRD-K16485616	core candidate	yes	no	Primary computational hypothesis
TC-H-106	BRD-K45044657	exploratory / original-method-sensitive	yes	no	Limited HiQ coverage; close training neighbor
NCH-51	BRD-K52522949	secondary candidate	yes	no	Measured support; scaffold-exposure caveat
Belinostat	BRD-K17743125	class support	yes	no	Measured HDAC-class support
PCI-24781	BRD-K12867552	class support	yes	no	Measured HDAC-class support
Panobinostat	BRD-K02130563	class support	yes	no	Measured HDAC-class support
Entinostat	BRD-K77908580	reference control	yes	no	Established HDAC reference
Vorinostat	BRD-K81418486	reference control	yes	no	Established HDAC reference
RG2833	BRD-K02389548	prediction only	yes	no	No measured LINCS profile
Tianeptinaline / BG-1010	BRD-A73972952	identity conflict; excluded	no	yes	Name–structure conflict; exact training exposure

Table S8: Official LINCS condition and quality coverage. “All cells” and “HiQ cells” are deliberately separated.

Candidate	All	QC	HiQ	All cells	HiQ cells	Times h	Dose levels	Min μM	Max μM
Mocetinostat	14	8	8	12	8	6 | 24	1	10.000	10.000
NCH-51	17	8	8	10	8	6 | 24	1	10.000	10.000
TC-H-106	6	4	4	5	4	6 | 24	1	10.000	10.000
Belinostat	24	13	12	12	6	6 | 24	4	0.370	10.000
PCI-24781	19	13	13	13	10	6 | 24	1	10.000	10.000
Panobinostat	126	93	84	15	10	6 | 24	20	0.001	10.000
Entinostat	101	77	67	14	10	6 | 24	13	0.040	10.000
Vorinostat	899	521	487	54	35	6 | 24 | 48	38	0.001	30.000
Tianeptinaline/BG-1010	6	1	1	2	1	6 | 24	1	10.000	10.000

The official LINCS cell count and the HiQ-eligible official-cell count are separate fields in Table S8. This resolves the earlier ambiguity in which a single "Cells" column could refer to either the full official metadata or the primary HiQ analysis.

Table S9: Pre-correction versus corrected annotation-defined HDAC holdout. The pre-correction row is provenance only and is excluded from inference.

Version	Test profiles	Test structures	Test cells	Δ profiles	Δ structures
discarded_pre_fix	1,565	21	55	0	0
corrected_final	1,856	30	55	291	9

The correction added nine annotation-defined structures and 291 profiles. NCH-51 was removed from training and placed into the final test set. Table S10 provides the complete compound-level inventory rather than only an aggregate count.

Table S10: Complete 30-structure corrected annotation-defined HDAC holdout inventory.

BRD ID	Compound	Profiles	Cells	Annotation source	Added by fix	NCH-51 removed
BRD-K85133207	HDAC1-selective	50	11	metadata	yes	no
BRD-K29313308	HDAC3-selective	55	11	metadata	yes	no
BRD-K76698671	HNHA	8	7	metadata	yes	no
BRD-K69840642	ISOX	48	43	metadata	yes	no
BRD-K83837640	JNJ-26481585	11	9	metadata_and_audit	no	no
BRD-K45528773	M-344	6	3	audit	no	no
BRD-K04466929	Merck60	51	8	metadata_and_audit	no	no
BRD-K52522949	NCH-51	17	11	metadata	yes	yes
BRD-K13169950	NSC-3852	20	14	metadata_and_audit	no	no
BRD-K12867552	PCI-24781	19	13	metadata_and_audit	no	no
BRD-K88742110	PCI-34051	43	11	metadata_and_audit	no	no
BRD-K45044657	TC-H-106	6	5	audit	no	no
BRD-K74761218	WT-171	15	11	metadata	yes	no
BRD-A63989068	apicidin	6	3	metadata	yes	no
BRD-K64606589	apicidin	85	14	metadata	yes	no
BRD-K17743125	belinostat	24	12	metadata_and_audit	no	no
BRD-K56957086	dacinostat	27	12	metadata_and_audit	no	no
BRD-K11558771	droxinostat	9	8	metadata_and_audit	no	no
BRD-K77908580	entinostat	101	14	metadata_and_audit	no	no
BRD-K13810148	givinostat	5	4	metadata_and_audit	no	no
BRD-K16485616	mocetinostat	14	12	metadata_and_audit	no	no
BRD-K02130563	panobinostat	126	15	metadata_and_audit	no	no
BRD-K67102207	phenylbutyrate	6	6	audit	no	no
BRD-K11663430	pyroxamide	13	10	audit	no	no
BRD-K22503835	scriptaid	38	14	metadata_and_audit	no	no
BRD-K52313696	tacedinaline	83	11	audit	no	no
BRD-K98426715	tubacin	7	6	metadata	yes	no
BRD-K00627859	tubastatin-a	51	8	metadata_and_audit	no	no
BRD-K41260949	valproic-acid	13	6	metadata_and_audit	no	no
BRD-K81418486	vorinostat	899	55	metadata_and_audit	no	no

Table S11: Strict leave-drug candidate membership.

Candidate	Train	Test	Test cells	Strict LDO
Belinostat	24	0	0	no
Entinostat	101	0	0	no
Mocetinostat	0	14	12	yes
NCH-51	17	0	0	no
PCI-24781	0	19	13	yes
Panobinostat	126	0	0	no
RG2833	0	0	0	no
TC-H-106	6	0	0	no
Tianeptinaline/BG-1010	6	0	0	no
Vorinostat	899	0	0	no

Supplementary Table S11 (continued). Strict leave-drug candidate metrics. Only Mocetinostat and PCI-24781 were true test candidates.

Candidate	Model	Seeds	Profiles	Cells	Pearson	95% low	95% high	Spearman
Mocetinostat	dual_stream	3	14	12	0.483	0.393	0.574	0.476
Mocetinostat	fingerprint_mlp	3	14	12	0.504	0.468	0.539	0.499
PCI-24781	dual_stream	3	19	13	0.308	0.152	0.464	0.270
PCI-24781	fingerprint_mlp	3	19	13	0.540	0.423	0.657	0.493

Only Mocetinostat and PCI-24781 satisfy strict leave-drug candidate membership. For PCI-24781, the seed-paired 95% interval favored the strong conventional fingerprint MLP comparator for profile correlations; no strict candidate result favored the dual stream. These three-seed intervals are uncertainty summaries rather than population-level significance tests.

4 Measured LINCS reversal and prediction–measurement concordance

Measured profiles were aggregated hierarchically: median within candidate–dose–time–official-cell condition, median across conditions within official cell, comparison with each cancer disease vector, and summary across official cells and cancers. The HiQ stratum is primary; QC-pass and all-profile rows are sensitivity analyses.

Table S12: Corrected-HDAC predicted versus HiQ measured reversal with 10,000 candidate–cancer crossed bootstrap iterations (seed 20260719).

Metric	Model	Candidates	Cancers	Pearson	95% low	95% high	Spearman	95% low	95% high
signed_wtrs	dual_stream	8	22	0.642	0.365	0.838	0.640	0.297	0.840
signed_wtrs	fingerprint_mlp	8	22	0.805	0.601	0.911	0.803	0.554	0.919
spearman_reversal	dual_stream	8	22	0.813	0.634	0.919	0.801	0.594	0.909
spearman_reversal	fingerprint_mlp	8	22	0.853	0.710	0.921	0.830	0.653	0.910

The crossed bootstrap independently resampled eight candidates and 22 cancers. This is more conservative than treating 176 candidate–cancer pairs as independent. Leave-one-candidate-out and leave-one-cancer-out influence tables are in Additional file 2 and retain positive associations. Because no condition-matched non-HDAC comparator set was available, these results quantify predicted–measured concordance for the eight compounds only; they do not establish preferential validation relative to alternative perturbagens.

Table S13: Primary official-condition HiQ measured reversal across 22 cancer signatures.

Candidate	Metric	HiQ	HiQ cells	Median	Q1	Q3	Positive cancers %
Belinostat	signed_wtrs	12	6	0.140	0.105	0.200	95.5
Belinostat	spearman_reversal	12	6	0.058	0.037	0.092	95.5
Entinostat	signed_wtrs	67	10	0.082	0.041	0.101	95.5
Entinostat	spearman_reversal	67	10	0.082	0.038	0.100	95.5
Mocetinostat	signed_wtrs	8	8	0.262	0.172	0.316	95.5
Mocetinostat	spearman_reversal	8	8	0.137	0.088	0.151	95.5
NCH-51	signed_wtrs	8	8	0.194	0.144	0.249	95.5
NCH-51	spearman_reversal	8	8	0.103	0.068	0.133	95.5
PCI-24781	signed_wtrs	13	10	0.139	0.111	0.211	95.5
PCI-24781	spearman_reversal	13	10	0.075	0.057	0.107	95.5
Panobinostat	signed_wtrs	84	10	0.191	0.152	0.224	95.5
Panobinostat	spearman_reversal	84	10	0.061	0.035	0.095	95.5
TC-H-106	signed_wtrs	4	4	0.119	0.081	0.160	90.9
TC-H-106	spearman_reversal	4	4	0.113	0.071	0.158	90.9
Vorinostat	signed_wtrs	487	35	0.054	0.036	0.122	95.5
Vorinostat	spearman_reversal	487	35	0.043	0.024	0.080	95.5

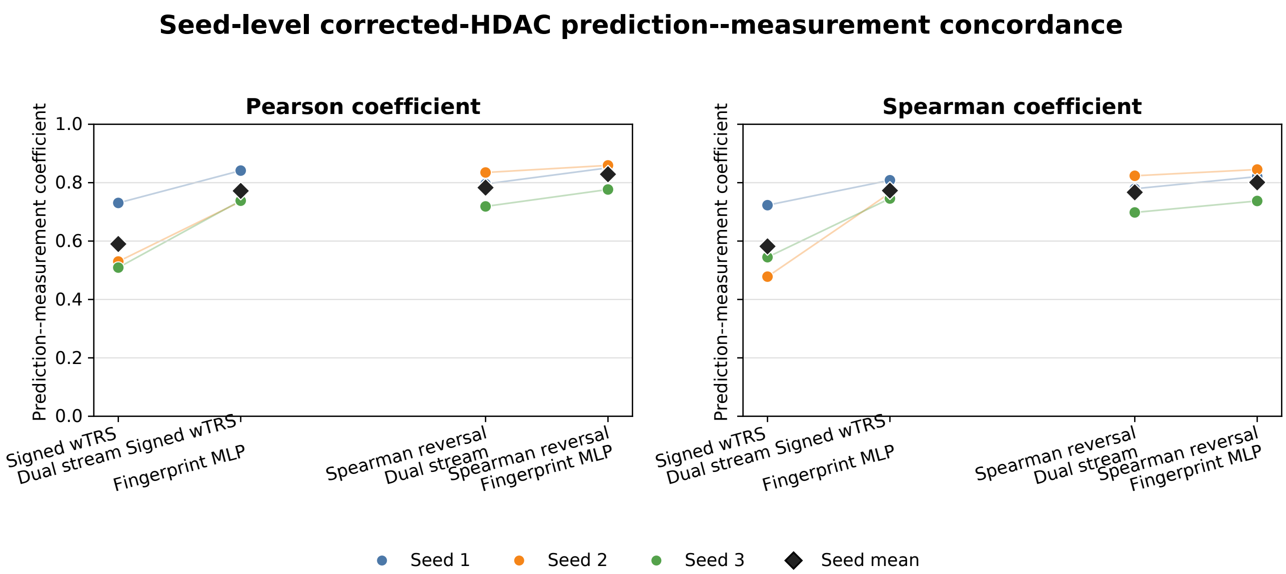


Figure S1: **Seed-level prediction–measurement concordance.** Pearson and Spearman coefficients are shown separately for every corrected annotation-defined HDAC seed, model, and primary reversal definition. Large markers denote seed means. The figure exposes run-to-run variability that is summarized by the crossed candidate–cancer intervals in the main text; it is not a second independent validation dataset.

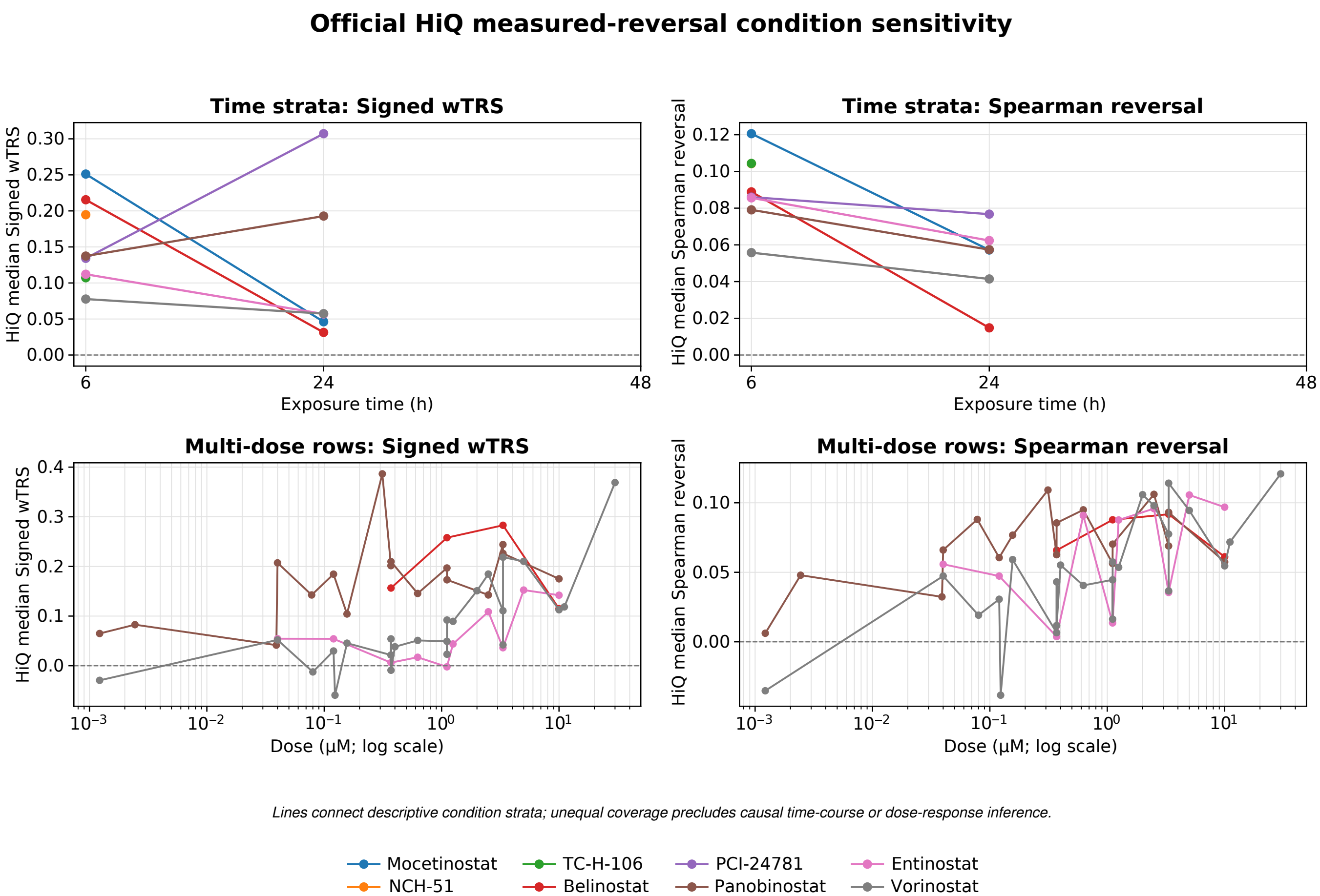


Figure S2: **Measured LINCS time- and dose-stratum sensitivity.** HiQ pan-cancer median signed wTRS and Spearman reversal are shown by exposure time for the eight primary compounds and by dose for the four reference or class-support compounds with more than one profiled dose. Unequal cells, signatures, and condition coverage prevent causal dose–response or time-course interpretation; these panels are descriptive sensitivity checks.

All eight primary compounds had positive median measured reversal under signed and rank-based definitions. These are experimental expression responses, not direct drug-sensitivity, target-engagement, or efficacy measurements. The all-profile comparison against 11,445 canonical compounds in Additional file 2 is unmatched for dose, time, cell, and quality and is therefore not a condition-matched non-HDAC null.

5 Candidate stability, structural exposure, and formal class enrichment

The primary stability hierarchy contains 48 configurations: four generalization or distribution-shift regimes, two models, three seeds, and two primary reversal metrics. Adding the legacy reversal-only magnitude score produces 72 configurations and is explicitly sensitivity analysis. These regime labels do not imply that every named candidate is unseen under every configuration. Candidate identity and training exposure override descriptive rank; this is why the high-ranking identity-conflicted Tianeptinaline/BG-1010 row remains excluded.

Table S14: Candidate stability under the 48-configuration primary hierarchy and 72-configuration legacy-inclusive sensitivity analysis.

Candidate	Role	Primary n	Primary median	Primary worst	Sensitivity n	Sensitivity median	Δ primary–sens.
Belinostat	class_support	48	70.0	56.8	72	92.8	-22.8
Entinostat	reference_control	48	77.9	16.3	72	66.2	11.7
Mocetinostat	core_candidate	48	92.2	53.8	72	91.9	0.3
NCH-51	secondary_candidate	48	88.5	65.6	72	95.4	-6.9
PCI-24781	class_support	48	57.6	32.3	72	57.6	0.0
Panobinostat	class_support	48	60.6	33.5	72	94.2	-33.5
RG2833	prediction_only	48	72.0	47.4	72	57.3	14.7
TC-H-106	original_method_sensitive	48	84.3	49.1	72	74.7	9.6
Tianeptinaline/BG-1010	identity_conflict_excluded	48	90.0	71.0	72	76.7	13.3
Vorinostat	reference_control	48	69.5	19.0	72	76.7	-7.2

Table S15: Corrected annotation-defined HDAC structural-neighbor and scaffold exposure audit.

Candidate	Role	Train	Test	Max Tanimoto	Top-10 mean	Exact train	Scaffold train	Same-scaffold n	Conflict
Mocetinostat	core candidate	0	14	0.537	0.447	no	no	0	no
TC-H-106	exploratory / original-method-sensitive	0	6	0.789	0.438	no	no	0	no
NCH-51	secondary candidate	0	17	0.431	0.366	no	yes	1	no
Belinostat	class support	0	24	0.500	0.339	no	yes	8	no
PCI-24781	class support	0	19	0.410	0.317	no	no	0	no
Panobinostat	class support	0	126	0.381	0.315	no	yes	1	no
Entinostat	reference control	0	101	0.558	0.429	no	no	0	no
Vorinostat	reference control	0	899	0.697	0.503	no	yes	296	no
RG2833	prediction only	0	0	0.574	0.407	no	no	0	no
Tianeptinaline/BG-1010	identity conflict; excluded	6	0	1.000	0.508	yes	yes	1	yes

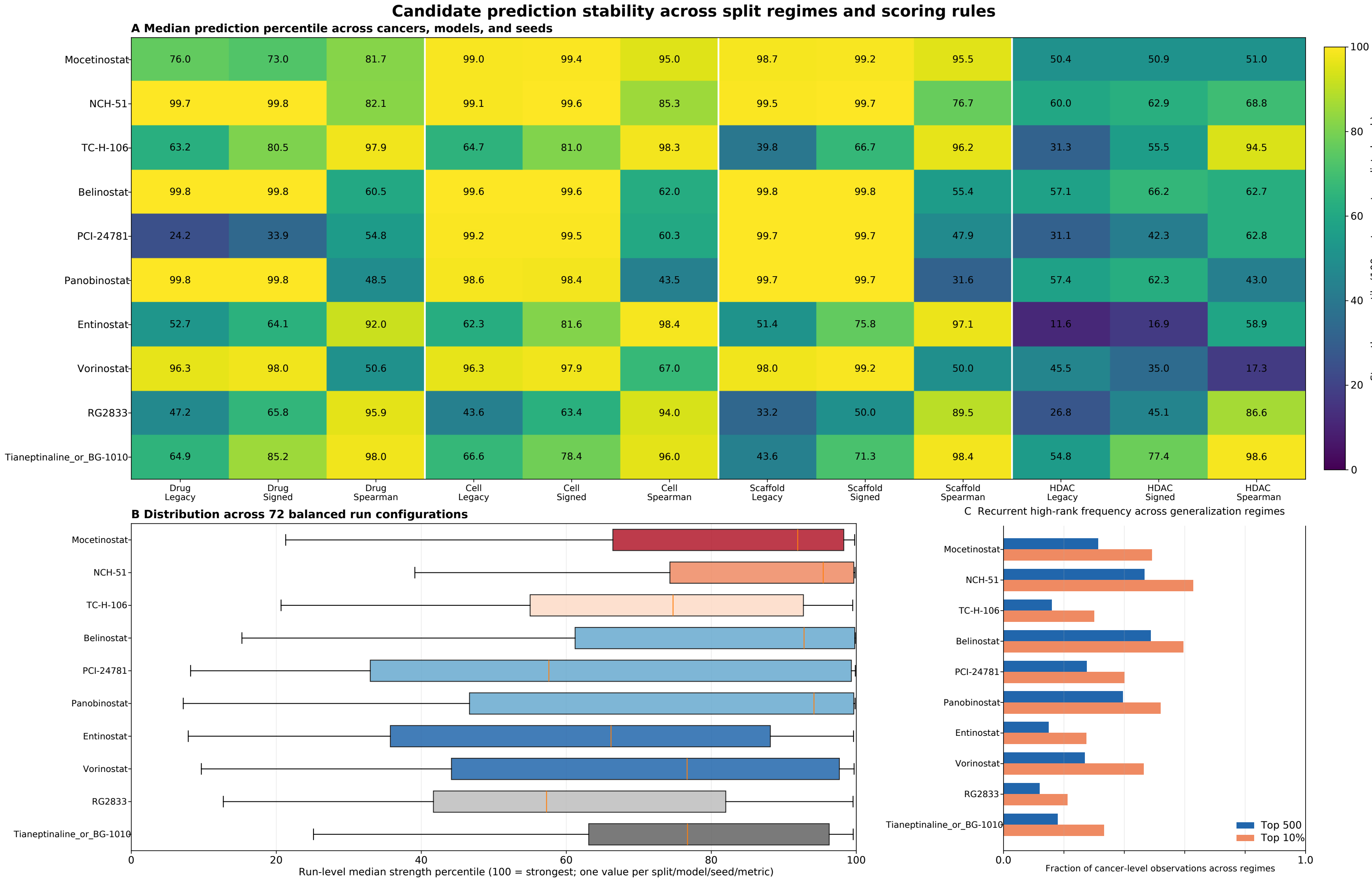


Figure S3: **Expanded split- and metric-level candidate stability sensitivity.** The heatmap reports the full 72-configuration legacy-inclusive audit by split and reversal definition, with run-level distributions and recurrent high-rank frequencies. This expands, rather than repeats, the 48-configuration primary summary in main Figure 4. Legacy wTRS remains a sensitivity metric and does not determine candidate eligibility.

Formal testing shows that the explicitly annotated class I subset is enriched under signed wTRS at every fixed revision cutoff, but not under Spearman reversal. For each metric, compounds were ordered by the mean of 528 equally weighted within-library rank fractions (24 screening runs across four generalization regimes, two models, and three seeds, multiplied by 22 cancers); 0 denotes the strongest rank, so lower mean rank fraction is better. Candidate stability uses a distinct strength-percentile orientation, where 100 denotes the strongest rank and higher is better. The cutoffs were fixed for the revision analysis but were not prospectively preregistered. The result is therefore metric-dependent rather than universal. The retained title refers to computational prioritization and does not imply model superiority, efficacy, or metric-independent enrichment. The broader HDAC definition and legacy metric are included in Additional file 2.

Table S16: Formal HDAC enrichment at fixed revision library cutoffs. Consensus ranks equally average 24 screening runs across 22 cancers per compound as mean rank fractions (0 = strongest); Fisher tests are one-sided and FDR is within each metric and annotation definition.

Metric	Definition	Top fraction	Top k	Library HDAC	Observed	Expected	Fold	*P*	FDR
signed_wtrs	annotation_defined_HDAC	0.5	143	39	4	0.20	20.42	4.37e-05	5.83e-05
signed_wtrs	annotation_defined_HDAC	1.0	285	39	5	0.39	12.81	4.22e-05	5.83e-05
signed_wtrs	annotation_defined_HDAC	5.0	1,424	39	11	1.95	5.64	2.16e-06	8.62e-06
signed_wtrs	annotation_defined_HDAC	10.0	2,848	39	13	3.90	3.33	6.48e-05	6.48e-05
signed_wtrs	explicit_class_I_target	0.5	143	26	4	0.13	30.64	8.36e-06	8.36e-06
signed_wtrs	explicit_class_I_target	1.0	285	26	5	0.26	19.22	5.37e-06	7.16e-06
signed_wtrs	explicit_class_I_target	5.0	1,424	26	11	1.30	8.46	1.81e-08	7.25e-08
signed_wtrs	explicit_class_I_target	10.0	2,848	26	12	2.60	4.61	2.47e-06	4.93e-06
spearman_reversal	annotation_defined_HDAC	0.5	143	39	1	0.20	5.11	1.78e-01	3.50e-01
spearman_reversal	annotation_defined_HDAC	1.0	285	39	2	0.39	5.12	5.80e-02	2.32e-01
spearman_reversal	annotation_defined_HDAC	5.0	1,424	39	3	1.95	1.54	3.09e-01	3.50e-01
spearman_reversal	annotation_defined_HDAC	10.0	2,848	39	5	3.90	1.28	3.50e-01	3.50e-01
spearman_reversal	explicit_class_I_target	0.5	143	26	1	0.13	7.66	1.23e-01	4.60e-01
spearman_reversal	explicit_class_I_target	1.0	285	26	1	0.26	3.84	2.30e-01	4.60e-01
spearman_reversal	explicit_class_I_target	5.0	1,424	26	2	1.30	1.54	3.76e-01	4.90e-01
spearman_reversal	explicit_class_I_target	10.0	2,848	26	3	2.60	1.15	4.90e-01	4.90e-01

Signed wTRS is linear in the perturbation vector and therefore sensitive to response magnitude, whereas Spearman reversal is invariant to positive rescaling. Table S16 supports class I HDAC enrichment only under signed wTRS; the non-significant Spearman result precludes a general enrichment claim across reversal definitions and leaves open a contribution from perturbational amplitude.

6 Frozen reversal-associated genes, pathways, and networks

Candidate roles were frozen before these analyses. At the 70% primary consensus threshold, the top 200 reversal-associated genes per candidate/control were carried forward. The term "reversal-associated" is used because gene selection derives from the same disease–perturbation opposition used for ranking; these genes are not direct pharmacological targets.

Table S17: Frozen reversal-associated gene definition and mapping coverage. Selection precedes network and pathway interpretation.

Candidate	Role	Threshold	Eligible	Stable direction	Top selected	STRING mappable	Mapping %
Mocetinostat	core_candidate	70.0	1,212	1,212	200	200	100.0
NCH-51	secondary_candidate	70.0	1,242	1,242	200	200	100.0
TC-H-106	exploratory_candidate	70.0	1,324	1,324	200	200	100.0
Belinostat	class_support	70.0	1,167	1,167	200	200	100.0
Entinostat	reference_control	70.0	880	880	200	200	100.0
Vorinostat	reference_control	70.0	1,035	1,035	200	200	100.0

Table S18: Leading 20 reversal-associated genes per candidate or control by frozen consensus rank. These are not direct drug targets.

Rank	Mocetinostat	NCH-51	TC-H-106	Belinostat	Entinostat	Vorinostat
1	BIRC5	UBE2C	ADH1B	UBE2C	PDK4	ATP1B2
2	TCF21	TCF21	CES1	RGN	AKAP12	FBLN5
3	LYVE1	GPX3	BIRC5	ATP1B2	TEK	CDT1
4	FOSB	FOSB	CHRDL1	CDT1	XPNPEP2	MCM2
5	TEK	LYVE1	UBE2C	NR4A3	FBLN5	LMNB2
6	NR4A3	ATP1B2	MYLK	PTGDS	CDK4	GPX3
7	GINS2	MCM2	PDK4	MCM2	FOXM1	CLN6
8	RAD51	AOX1	AOX1	FBLN5	CDCA8	TCF21
9	PLK1	LIFR	GINS2	POLE2	FAM107A	FOXM1
10	CCNA2	CCDC69	CDT1	CDK4	LEPR	CCNB2
11	GPM6A	RGN	ABCA8	LMNB2	RAD51AP1	KIF4A
12	HBB	PDGFRA	CLEC3B	TRIB3	CCDC86	AKAP12
13	BUB1	TXNIP	PDGFRA	DNAJB4	FEN1	TTK
14	MCM10	CENPM	FOSB	ZNF331	DNAJC9	NCAPH
15	HJURP	CDT1	AQP7	SLC25A20	ADH1B	ZNF282
16	TPX2	CCNB2	PRELP	CCNB2	BIRC5	SPC25
17	CDCA3	KIF20A	AKAP12	PLK1	UBE2C	TRIB3
18	ASF1B	PLK1	AOC3	FOXM1	CHRDL1	ECT2
19	KCNK3	CCNA2	FMO2	CCNA2	CES1	PLK1
20	CKAP2	FOXM1	PRKAR2B	GTSE1	GPX3	CDK4

Table S19: Scope of significant g:Profiler results by candidate, reversal direction, and ontology/source.

Candidate	Gene-set scope	Source	Significant terms
Belinostat	all_selected	GO:BP	167
Belinostat	all_selected	GO:CC	44
Belinostat	all_selected	GO:MF	4
Belinostat	all_selected	KEGG	2
Belinostat	all_selected	REAC	50
Belinostat	disease_down_drug_up	GO:BP	7
Belinostat	disease_up_drug_down	GO:BP	272
Belinostat	disease_up_drug_down	GO:CC	75
Belinostat	disease_up_drug_down	GO:MF	68
Belinostat	disease_up_drug_down	KEGG	7
Belinostat	disease_up_drug_down	REAC	100
Entinostat	all_selected	GO:BP	182
Entinostat	all_selected	GO:CC	56
Entinostat	all_selected	GO:MF	31
Entinostat	all_selected	KEGG	5
Entinostat	all_selected	REAC	93
Entinostat	disease_down_drug_up	GO:BP	7
Entinostat	disease_down_drug_up	GO:CC	17
Entinostat	disease_down_drug_up	KEGG	1
Entinostat	disease_down_drug_up	REAC	1
Entinostat	disease_up_drug_down	GO:BP	281
Entinostat	disease_up_drug_down	GO:CC	111
Entinostat	disease_up_drug_down	GO:MF	106
Entinostat	disease_up_drug_down	KEGG	9
Entinostat	disease_up_drug_down	REAC	146
Mocetinostat	all_selected	GO:BP	163
Mocetinostat	all_selected	GO:CC	30
Mocetinostat	all_selected	GO:MF	2
Mocetinostat	all_selected	KEGG	2
Mocetinostat	all_selected	REAC	51
Mocetinostat	disease_down_drug_up	GO:BP	88
Mocetinostat	disease_down_drug_up	GO:CC	13
Mocetinostat	disease_down_drug_up	REAC	9
Mocetinostat	disease_up_drug_down	GO:BP	272
Mocetinostat	disease_up_drug_down	GO:CC	88
Mocetinostat	disease_up_drug_down	GO:MF	30
Mocetinostat	disease_up_drug_down	KEGG	7
Mocetinostat	disease_up_drug_down	REAC	178
NCH-51	all_selected	GO:BP	173
NCH-51	all_selected	GO:CC	44
NCH-51	all_selected	GO:MF	4
NCH-51	all_selected	KEGG	4
NCH-51	all_selected	REAC	47
NCH-51	disease_down_drug_up	GO:CC	3
NCH-51	disease_down_drug_up	KEGG	1
NCH-51	disease_down_drug_up	REAC	1
NCH-51	disease_up_drug_down	GO:BP	302
NCH-51	disease_up_drug_down	GO:CC	91
NCH-51	disease_up_drug_down	GO:MF	69
NCH-51	disease_up_drug_down	KEGG	11
NCH-51	disease_up_drug_down	REAC	102
TC-H-106	all_selected	GO:BP	12
TC-H-106	all_selected	GO:CC	36
TC-H-106	all_selected	REAC	4
TC-H-106	disease_down_drug_up	GO:BP	21
TC-H-106	disease_down_drug_up	GO:CC	13
TC-H-106	disease_down_drug_up	KEGG	1
TC-H-106	disease_up_drug_down	GO:BP	181
TC-H-106	disease_up_drug_down	GO:CC	83
TC-H-106	disease_up_drug_down	GO:MF	3
TC-H-106	disease_up_drug_down	KEGG	3
TC-H-106	disease_up_drug_down	REAC	66
Vorinostat	all_selected	GO:BP	211
Vorinostat	all_selected	GO:CC	69
Vorinostat	all_selected	GO:MF	42
Vorinostat	all_selected	KEGG	5
Vorinostat	all_selected	REAC	61
Vorinostat	disease_down_drug_up	GO:BP	12
Vorinostat	disease_down_drug_up	GO:MF	9
Vorinostat	disease_up_drug_down	GO:BP	288
Vorinostat	disease_up_drug_down	GO:CC	93
Vorinostat	disease_up_drug_down	GO:MF	61
Vorinostat	disease_up_drug_down	KEGG	12
Vorinostat	disease_up_drug_down	REAC	124

Supplementary Table S19 (continued). Representative nonredundant enrichment terms after corrected gene-intersection redundancy filtering. Probabilities are shown in scientific notation rather than rounded to zero.

Candidate	Gene-set scope	Source	Term ID	Nonredundant term	P_{FDR}	Overlap	Term size	Effective domain
Belinostat	disease_up_drug_down	GO:BP	GO:0007059	chromosome segregation	1.61e-29	36	294	11,154
Belinostat	disease_up_drug_down	GO:BP	GO:0010965	regulation of mitotic sister chromatid separation	2.36e-18	15	47	11,154
Belinostat	disease_up_drug_down	GO:CC	GO:0098687	chromosomal region	1.51e-16	25	283	11,154
Belinostat	disease_up_drug_down	GO:CC	GO:0005819	spindle	2.68e-14	23	305	11,154
Belinostat	disease_up_drug_down	GO:MF	GO:0008017	microtubule binding	1.45e-07	14	182	11,154
Belinostat	disease_up_drug_down	GO:MF	GO:0120543	macromolecular conformation isomerase activity	3.75e-05	12	217	11,154
Belinostat	disease_up_drug_down	KEGG	KEGG:04110	Cell cycle	3.28e-10	14	130	11,154
Belinostat	disease_up_drug_down	KEGG	KEGG:03030	DNA replication	4.93e-07	7	34	11,154
Belinostat	disease_up_drug_down	REAC	REAC:R-HSA-1640170	Cell Cycle	1.29e-30	43	505	11,154
Belinostat	disease_up_drug_down	REAC	REAC:R-HSA-69620	Cell Cycle Checkpoints	4.53e-15	21	209	11,154
Entinostat	disease_up_drug_down	GO:BP	GO:1903047	mitotic cell cycle process	1.21e-16	37	568	11,154
Entinostat	disease_up_drug_down	GO:BP	GO:0022613	ribonucleoprotein complex biogenesis	1.77e-13	26	333	11,154
Entinostat	disease_up_drug_down	GO:CC	GO:0098687	chromosomal region	1.94e-18	29	283	11,154
Entinostat	disease_up_drug_down	GO:CC	GO:0000940	outer kinetochore	5.51e-10	7	13	11,154
Entinostat	disease_up_drug_down	GO:MF	GO:0003723	RNA binding	3.72e-12	44	1,188	11,154
Entinostat	disease_up_drug_down	GO:MF	GO:0140640	catalytic activity, acting on a nucleic acid	3.72e-12	27	410	11,154
Entinostat	disease_up_drug_down	KEGG	KEGG:04110	Cell cycle	6.12e-11	16	130	11,154
Entinostat	disease_up_drug_down	KEGG	KEGG:03030	DNA replication	1.41e-07	8	34	11,154
Entinostat	disease_up_drug_down	REAC	REAC:R-HSA-1640170	Cell Cycle	2.80e-24	42	505	11,154
Entinostat	disease_up_drug_down	REAC	REAC:R-HSA-69306	DNA Replication	7.78e-11	15	111	11,154
Mocetinostat	disease_up_drug_down	GO:BP	GO:1903047	mitotic cell cycle process	7.66e-30	41	568	11,154
Mocetinostat	disease_up_drug_down	GO:BP	GO:0000070	mitotic sister chromatid segregation	7.46e-18	19	142	11,154
Mocetinostat	disease_up_drug_down	GO:CC	GO:0000793	condensed chromosome	1.12e-14	19	197	11,154
Mocetinostat	disease_up_drug_down	GO:CC	GO:0005819	spindle	4.62e-13	20	305	11,154
Mocetinostat	disease_up_drug_down	GO:MF	GO:0008017	microtubule binding	1.34e-05	11	182	11,154
Mocetinostat	disease_up_drug_down	GO:MF	GO:0016853	isomerase activity	4.59e-03	10	324	11,154
Mocetinostat	disease_up_drug_down	KEGG	KEGG:04110	Cell cycle	1.09e-06	10	130	11,154
Mocetinostat	disease_up_drug_down	KEGG	KEGG:04114	Oocyte meiosis	9.33e-04	6	95	11,154
Mocetinostat	disease_up_drug_down	REAC	REAC:R-HSA-69278	Cell Cycle, Mitotic	4.71e-25	33	405	11,154
Mocetinostat	disease_up_drug_down	REAC	REAC:R-HSA-69618	Mitotic Spindle Checkpoint	3.78e-11	12	86	11,154
NCH-51	disease_up_drug_down	GO:BP	GO:1903047	mitotic cell cycle process	5.48e-33	46	568	11,154
NCH-51	all_selected	GO:BP	GO:0007059	chromosome segregation	8.04e-19	38	294	11,154
NCH-51	disease_up_drug_down	GO:CC	GO:0000793	condensed chromosome	3.11e-17	22	197	11,154
NCH-51	disease_up_drug_down	GO:CC	GO:0005819	spindle	4.20e-16	24	305	11,154
NCH-51	disease_up_drug_down	GO:MF	GO:0008017	microtubule binding	5.17e-08	14	182	11,154
NCH-51	disease_up_drug_down	GO:MF	GO:0120543	macromolecular conformation isomerase activity	1.58e-05	12	217	11,154
NCH-51	disease_up_drug_down	KEGG	KEGG:04110	Cell cycle	2.06e-09	13	130	11,154
NCH-51	disease_up_drug_down	KEGG	KEGG:03030	DNA replication	2.85e-07	7	34	11,154
NCH-51	disease_up_drug_down	REAC	REAC:R-HSA-69278	Cell Cycle, Mitotic	2.13e-30	39	405	11,154
NCH-51	disease_up_drug_down	REAC	REAC:R-HSA-68886	M Phase	4.29e-11	19	279	11,154
TC-H-106	disease_up_drug_down	GO:BP	GO:0022613	ribonucleoprotein complex biogenesis	3.48e-12	22	333	11,154
TC-H-106	disease_up_drug_down	GO:BP	GO:0006260	DNA replication	1.39e-06	13	213	11,154
TC-H-106	disease_up_drug_down	GO:CC	GO:1990904	ribonucleoprotein complex	2.01e-07	19	510	11,154

Continued on next page

Table continued

Candidate	Gene-set scope	Source	Term ID	Nonredundant term	P_{FDR}	Overlap	Term size	Effective domain
TC-H-106	disease_up_drug_down	GO:CC	GO:0005730	nucleolus	6.61e-07	22	751	11,154
TC-H-106	disease_up_drug_down	GO:MF	GO:0003723	RNA binding	1.17e-08	32	1,188	11,154
TC-H-106	disease_up_drug_down	GO:MF	GO:0016887	ATP hydrolysis activity	1.12e-02	10	305	11,154
TC-H-106	disease_up_drug_down	KEGG	KEGG:03040	Spliceosome	1.60e-04	8	105	11,154
TC-H-106	disease_up_drug_down	KEGG	KEGG:03030	DNA replication	8.24e-03	4	34	11,154
TC-H-106	disease_up_drug_down	REAC	REAC:R-HSA-69278	Cell Cycle, Mitotic	2.53e-06	17	405	11,154
TC-H-106	disease_up_drug_down	REAC	REAC:R-HSA-8953854	Metabolism of RNA	1.19e-05	18	553	11,154
Vorinostat	disease_up_drug_down	GO:BP	GO:0007059	chromosome segregation	2.31e-25	36	294	11,154
Vorinostat	disease_up_drug_down	GO:BP	GO:0010965	regulation of mitotic sister chromatid separation	8.87e-17	15	47	11,154
Vorinostat	disease_up_drug_down	GO:CC	GO:0098687	chromosomal region	9.27e-22	32	283	11,154
Vorinostat	all_selected	GO:CC	GO:0005694	chromosome	7.93e-16	72	1,361	11,154
Vorinostat	disease_up_drug_down	GO:MF	GO:0120543	macromolecular conformation isomerase activity	1.88e-06	15	217	11,154
Vorinostat	disease_up_drug_down	GO:MF	GO:0140640	catalytic activity, acting on a nucleic acid	1.88e-06	20	410	11,154
Vorinostat	disease_up_drug_down	KEGG	KEGG:04110	Cell cycle	5.12e-11	16	130	11,154
Vorinostat	disease_up_drug_down	KEGG	KEGG:03030	DNA replication	2.99e-06	7	34	11,154
Vorinostat	disease_up_drug_down	REAC	REAC:R-HSA-1640170	Cell Cycle	4.00e-25	43	505	11,154
Vorinostat	disease_up_drug_down	REAC	REAC:R-HSA-2500257	Resolution of Sister Chromatid Cohesion	1.25e-11	15	98	11,154

The g:Profiler request submitted 11,144 unique measured Entrez identifiers as the custom background. The service response reported an effective mapped statistical domain of 11,154. These values answer different questions and are both retained. For every term, the returned per-query evidence array was aligned to the mapped-query order in the frozen response; nonempty entries identify the intersecting Entrez genes, while GO evidence codes are retained in a separate machine-readable field. All 5,037 reconstructed gene intersections match the service-reported intersection sizes. Table S19 reports candidate/source scope and de-duplicates representative term names; P values remain in scientific notation rather than appearing as "0.00." Additional file 2 contains the complete term table and a cache/hash audit.

Table S20: STRING physical and functional network scope at required score 700.

Candidate	Network	Selected	Nodes with edges	Edges	Density	Components	Largest component
Mocetinostat	physical	200	49	52	0.003	160	30
Mocetinostat	functional	200	110	577	0.029	102	66
NCH-51	physical	200	56	59	0.003	156	25
NCH-51	functional	200	116	944	0.047	98	86
TC-H-106	physical	200	43	52	0.003	167	13
TC-H-106	functional	200	88	243	0.012	121	65
Belinostat	physical	200	55	61	0.003	155	27
Belinostat	functional	200	105	968	0.049	104	81
Entinostat	physical	200	65	117	0.006	142	38
Entinostat	functional	200	118	673	0.034	92	59
Vorinostat	physical	200	63	106	0.005	147	31
Vorinostat	functional	200	122	918	0.046	87	95

Supplementary Table S20 (continued). Primary-candidate overlap with Entinostat and Vorinostat reversal-associated gene sets.

Candidate	Reference	Genes A	Genes B	Shared	Jaccard
Mocetinostat	Entinostat	200	200	86	0.274
Mocetinostat	Vorinostat	200	200	63	0.187
NCH-51	Entinostat	200	200	74	0.227
NCH-51	Vorinostat	200	200	100	0.333
TC-H-106	Entinostat	200	200	98	0.325
TC-H-106	Vorinostat	200	200	62	0.183

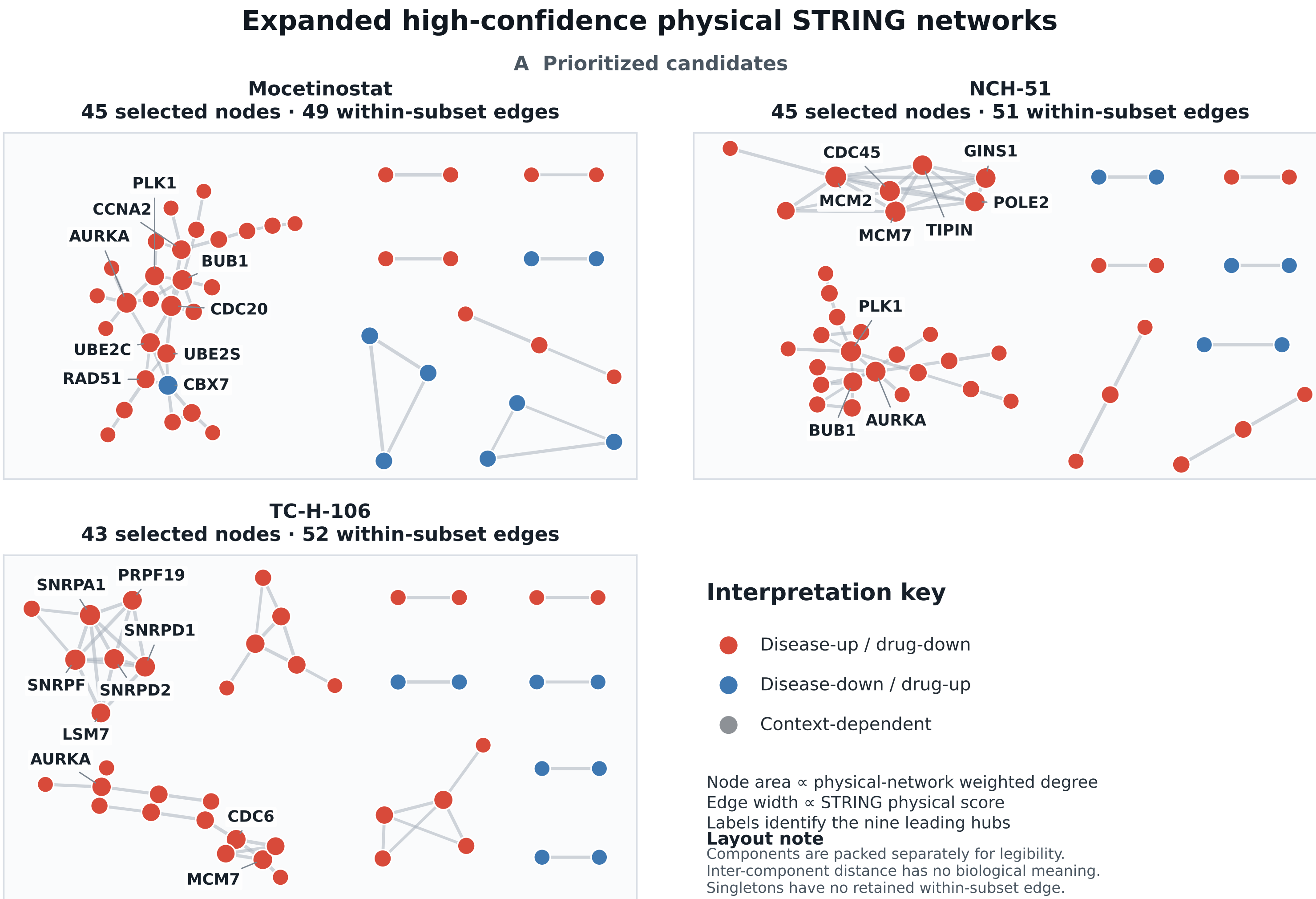
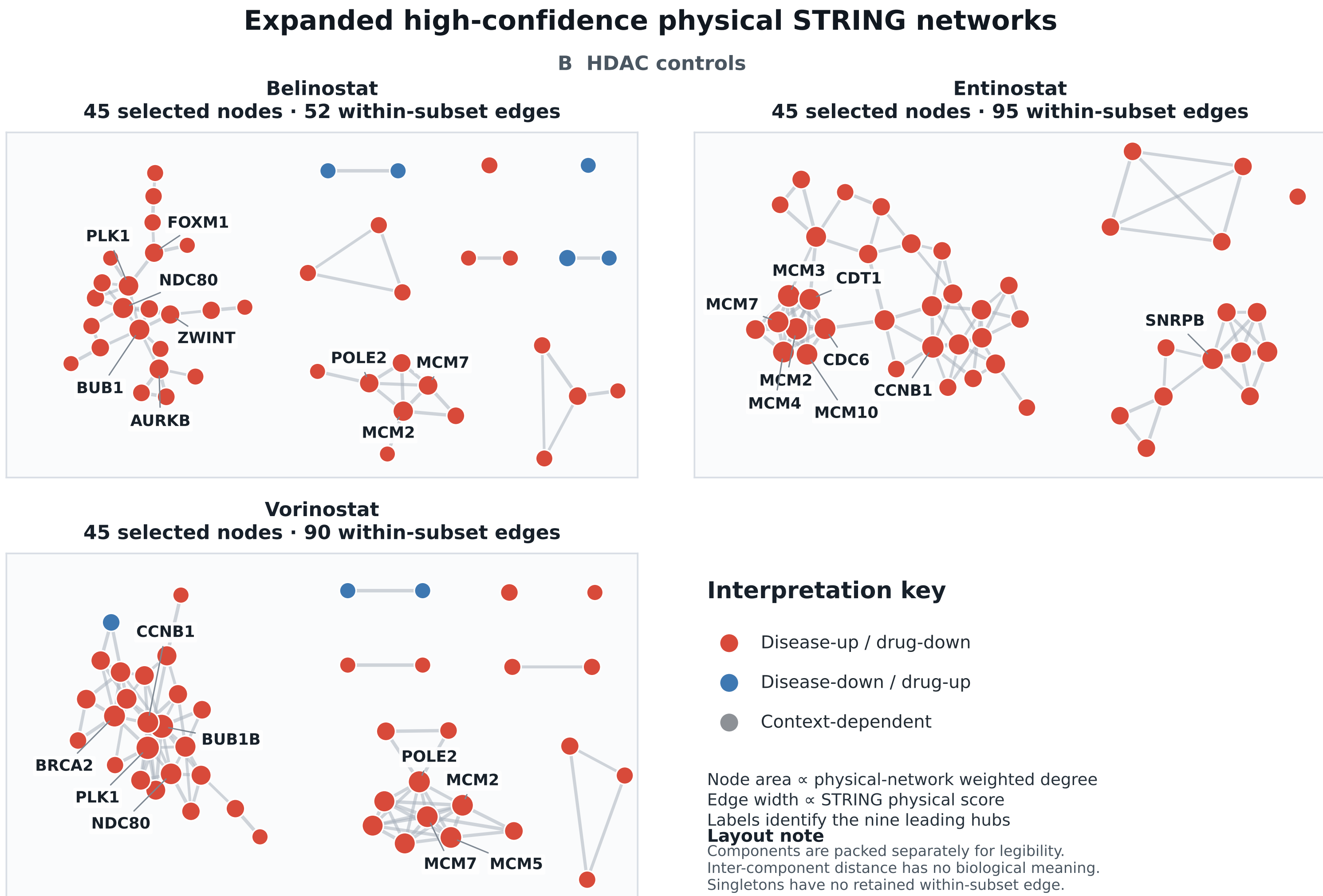


Figure S4: **(A) Expanded high-confidence physical STRING plotting subsets.** Mocetinostat, NCH-51, and TC-H-106 are shown with up to 45 non-isolated nodes selected by weighted degree. Leading hubs are labeled. Full 200-gene node tables and all physical/functional edges are in Additional file 2.



Supplementary Figure S4 (continued), panel B. Expanded high-confidence physical STRING plotting subsets for controls. Belinostat, Entinostat, and Vorinostat are shown under the same plotting policy. Nodes remain reversal-associated genes, not direct drug targets.

7 DepMap genetic-dependency context

Table S21: DepMap Public 26Q1 pan-cancer class I HDAC gene-effect summary.

Gene	Models	Median	95% low	95% high	≤ -0.5 %	≤ -1.0 %
HDAC1	1,208	-0.107	-0.118	-0.093	4.5	1.1
HDAC2	1,208	0.001	-0.008	0.013	3.1	0.5
HDAC3	1,208	-1.215	-1.249	-1.188	95.1	68.2
HDAC8	1,208	-0.121	-0.134	-0.107	4.9	0.1

Supplementary Table S21 (continued). Lineage heterogeneity. Effect sizes are modest despite FDR significance.

Gene	Lineages	Models	Kruskal H	P	ϵ^2	FDR
HDAC1	24	1,193	107.32	7.49e-13	0.072	1.50e-12
HDAC2	24	1,193	119.95	4.24e-15	0.083	1.70e-14
HDAC3	24	1,193	95.64	7.90e-11	0.062	1.05e-10
HDAC8	24	1,193	71.71	6.61e-07	0.042	6.61e-07

HDAC3 is the dominant pan-cancer dependency in DepMap Public 26Q1. The lineage tests are statistically significant with modest effect sizes, and pairwise HDAC correlations are weak. These results provide genetic-loss context but do not identify a candidate's pharmacological isoform, efficacy, or normal-tissue safety.

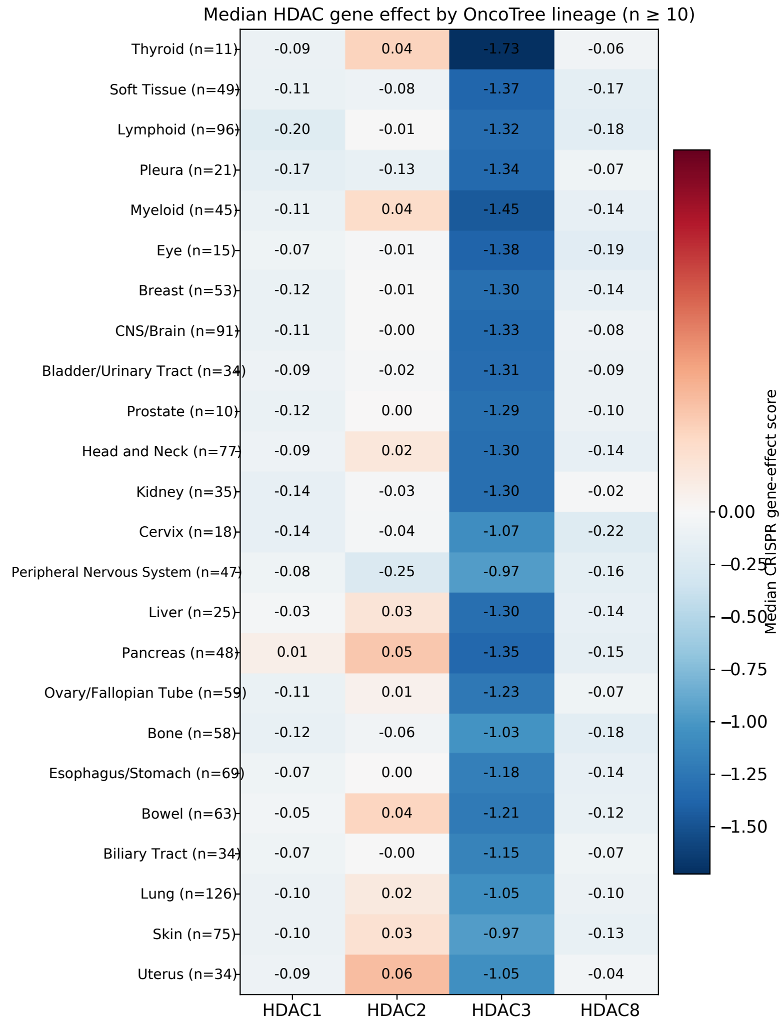


Figure S5: **DepMap lineage heterogeneity.** Median class I HDAC CRISPR gene effects are shown only for lineages meeting the prespecified minimum of ten models. More negative values indicate stronger loss-of-function effect.

8 Controlled docking and receptor sensitivity

Table S22: 4LXZ crystallographic redocking control. The prespecified RMSD gate was 2 Angstrom.

Method	Seeds	Seed coverage	Cluster poses	Median score	Medoid RMSD	Zn min	Zn max
vina_standard	3	3	8	-7.290	4.771	2.348	3.645
ad4zn	3	3	3	-7.491	1.091	2.123	3.108

Supplementary Table S22 (continued). Frozen 4LXZ control/candidate panel. Favorable score is not evidence of intended zinc geometry.

Compound	Role	3-seed	Poses	Cluster score	Medoid score	ZBG min	ZBG max	Nearest heteroatom
Vorinostat	crystallographic_positive_control	yes	3	-7.491	-7.532	2.123	3.108	2.123
Vorinostat O-methyl decoy	designed zinc-chelation sensitivity control	yes	6	-7.998	-7.791	6.799	8.992	4.078
Mocetinostat	core_candidate	yes	7	-11.024	-12.289	12.954	13.872	2.002
NCH-51	secondary_candidate	yes	8	-8.357	-9.094	–	–	2.448
TC-H-106	original_method_sensitive_candidate	yes	5	-9.674	-9.895	10.442	13.748	3.858
Entinostat	reference_control	yes	10	-10.703	-11.996	11.852	13.867	1.918
Belinostat	class_support_control	yes	3	-9.336	-9.383	8.918	9.525	2.841

Supplementary Table S22 (continued). HDAC2 4LXZ versus aligned HDAC1 4BKX receptor sensitivity.

Compound	Pose RMSD	4LXZ score	4BKX score	Δ score	4LXZ nearest Zn	4BKX nearest Zn
Vorinostat	8.357	-7.491	-8.111	-0.620	2.123	2.324
Vorinostat O-methyl decoy	1.872	-7.998	-7.767	0.231	4.078	2.210
Mocetinostat	1.080	-11.024	-10.864	0.160	2.002	2.176
NCH-51	1.870	-8.357	-8.457	-0.100	2.448	2.435
TC-H-106	1.508	-9.674	-8.886	0.788	3.858	3.895
Entinostat	1.280	-10.703	-10.539	0.164	1.918	1.907
Belinostat	8.787	-9.336	-7.915	1.421	2.841	2.140

The redocking control establishes protocol suitability only for one known cocrystal geometry under the zinc-aware potential. The O-methyl decoy demonstrates that a favorable score can coexist with an implausible prespecified zinc-binding-group distance. Several candidate poses change across 4LXZ and 4BKX. Consequently, docking is a protocol/receptor-sensitivity analysis with unresolved pose plausibility, not binding validation.

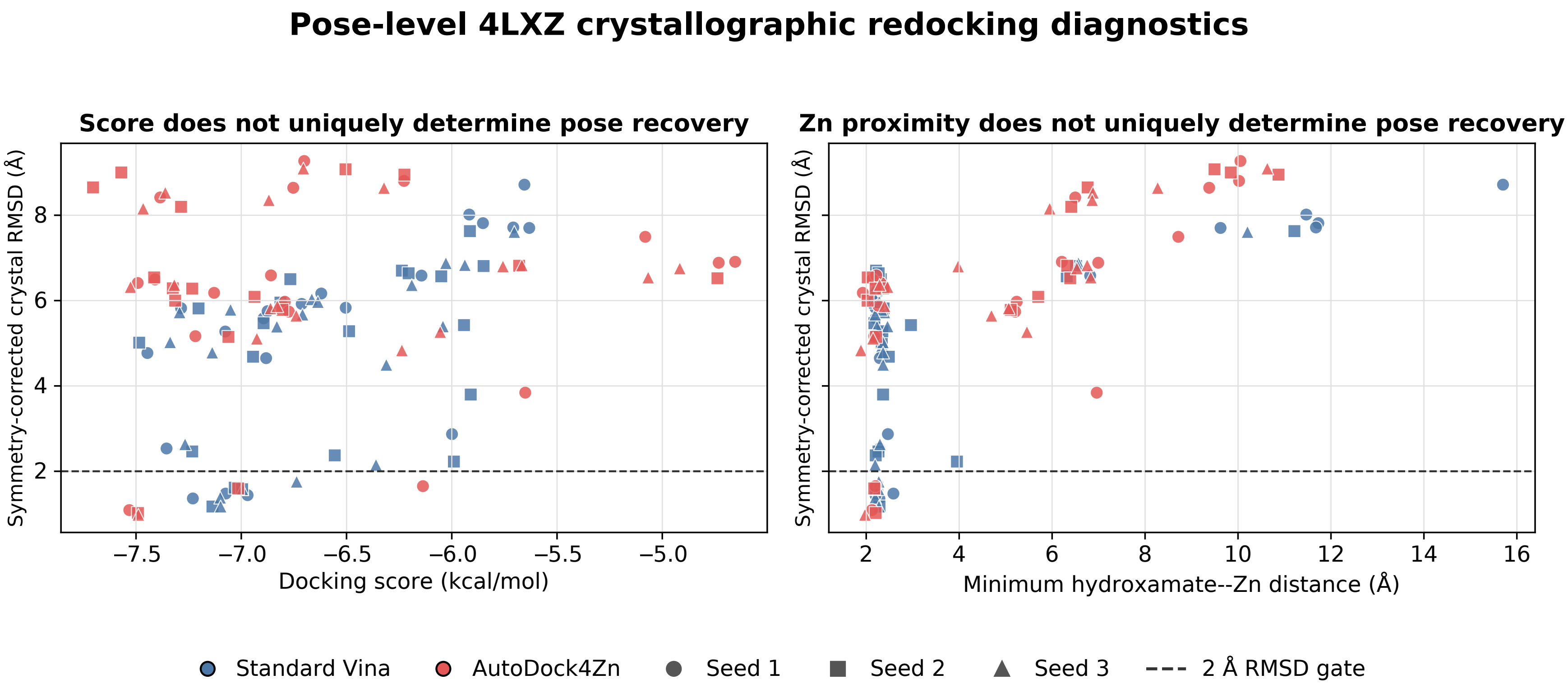


Figure S6: **Pose-level crystallographic redocking diagnostics.** Every retained pose from seeds 1–3 is shown by docking method. Score, symmetry-corrected heavy-atom RMSD, and hydroxamate–zinc distance are displayed jointly with the prespecified 2 Å RMSD gate. These pose-level diagnostics expand main Figure 9 and show why score or zinc proximity alone cannot establish a recovered binding geometry.

9 External drug-sensitivity availability and reproducibility

Table S23: External direct drug-sensitivity availability and null results. The public-transcriptomic option was completed; no surrogate is presented as direct validation.

Scope	Endpoint	Availability	Result	Interpretation
TC-H-106	Direct public viability response	Unavailable in frozen public-data audit	Not performed	No direct external drug-sensitivity validation claimed
Entinostat	GDSC surrogate correlation	Available but not candidate matched	Pearson R=-0.052; P=0.859	Null result; cannot validate TC-H-106
All frozen candidates	New wet-lab viability/biochemical assay	Not performed; reviewer suggested wet-lab assays or public transcriptomic validation	Official-condition measured LINCS used as the public transcriptomic option	Does not replace viability or biochemical validation

The non-significant Entinostat surrogate result is retained as a null result and cannot validate TC-H-106. No direct candidate-matched public viability response was found for TC-H-106 in the frozen audit. This limitation is stated in the manuscript and response letter rather than replaced with a post hoc proxy.

Table S24: Reproducibility inventory. Unrecorded CPU/RAM values are disclosed rather than guessed.

Item	Frozen value	Provenance/limitation
Neural hardware	NVIDIA GeForce RTX 4090	Recorded by run manifests
CPU model	Not captured in frozen AutoDL manifests	Not reconstructed or fabricated
RAM	Not captured in frozen AutoDL manifests	Not reconstructed or fabricated
Python	3.12.3	Run manifests
PyTorch/CUDA	2.5.1+cu124	Run manifests
RDKit	2025.03.6 for docking; modeling canonicalization version in repository environment	Environment manifests
Neural benchmark runtime	5,112.6 s summed across 34 reported runs	Run-level values in Table S5
Docking runtimes	330.0 s redocking; 2,121.9 s 4LXZ panel; 2,134.9 s 4BKX sensitivity	CPU workflows
Split seed	42	Top-level grouping
Run seeds	1–5 pair; 1–3 other splits	Frozen benchmark
Crossed-bootstrap seed	20260719	10,000 iterations
Repository evidence snapshot	fefbf2f4fa7399983d4d4041dcb8e7e91b84d17f	Corrected statistics, scripts, exclusions, and tracker on major-revision-2026

CPU model and RAM were not captured in the frozen AutoDL manifests. They are explicitly marked unrecorded rather than inferred from the current environment. The corrected evidence snapshot is archived at repository commit `fefbf2f4fa7399983d4d4041dcb8e7e91b84d17f`; later document-only commits do not alter that frozen analysis snapshot. The repository `REPRODUCIBILITY.md` file distinguishes original outputs, analyses regenerated during revision, retained derived evidence, large nonredistributed inputs or run artifacts, and unavailable intermediates.

10 Machine-readable companion

Additional file 2 contains the complete preserved run-level, candidate-level, network, pathway, DepMap, docking, sensitivity, and reproducibility tables with filters and a data dictionary. A TCGA cohort manifest identifies all 22 GDC projects and the frozen disease-matrix hashes; per-cohort final tumor/normal counts were not retained and are not retrospectively reconstructed. A separate condition-level manifest records the schema of the analysis-time 83,490-row table and explicitly discloses that the source file and its SHA256 were not preserved in the frozen portable package. All preserved aggregates used for the reported figures and inferences are included. These retained tables support direct verification of the reported statistics and recompilation of the documents from frozen figure assets, but a self-contained bitwise rerun of every training and screening stage from GitHub and current public inputs alone is not guaranteed because large checkpoints, prediction arrays, and specified analysis-time intermediates are absent from the portable archive. Repository-relative source paths are used; no AutoDL absolute paths are required to interpret the workbook. Blank fields are retained only when a measurement is not applicable or was not recorded, and the corresponding reason is supplied in a status or note column.